

● EASY

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Geometry and Trigonometry

08

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**D****D.** 136

Choice D is correct. In the figure shown, the angles with measures w° and z° are vertical angles. Since vertical angles are congruent, $w = z$. Therefore, if $w = 136$, the value of z is 136.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure, in degrees, of an angle that's supplementary, not congruent, to the angle with measure w° .

Choice C is incorrect and may result from conceptual or calculation errors.

Q2**A****A.** 22

Choice A is correct. It's given that BC is parallel to AD . Since line t intersects $\overline{BC}$ and $\overline{AD}$, it's a transversal. The angles in the figure marked as x° and y° are on opposite sides of the transversal and lie between the parallel segments, so they are alternate interior angles. When two parallel segments are intersected by a transversal, alternate interior angles have equal measure. It's also given that $x = 22$. Therefore, the value of y is 22.

Choice B is incorrect. This is the value of $2x$, not the value of y .

Choice C is incorrect. This is the complement of 22, not the value of y .

Choice D is incorrect. This is the supplement of 22, not the value of y .

Q3**C****C.** 60

Choice C is correct. It's given that triangle XYZ is obtained by a dilation of triangle ABC . It follows that triangle ABC is similar to triangle XYZ , where A corresponds to X . Since corresponding angles in similar triangles have the same measure and the measure of angle A is 60 degrees, it follows that the measure of angle X is also 60 degrees.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q4**B****B.** 72°

Choice B is correct. It's given that triangle ABC is congruent to triangle DEF . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that angle A corresponds to angle D , and that the measure of angle A is 18° . It's also given that the measures of angles B and E are 90° . Since these angles have equal measure, they are corresponding angles. It follows that angle C corresponds to angle F . Let x° represent the measure of angle C . Since the sum of the measures of the interior angles of a triangle is 180° , it follows that $18^\circ + 90^\circ + x^\circ = 180^\circ$, or $108^\circ + x^\circ = 180^\circ$. Subtracting 108° from both sides of this equation yields $x^\circ = 72^\circ$. Therefore, the measure of angle C is 72° . Since angle C corresponds to angle F , it follows that the measure of angle F is also 72° .

Choice A is incorrect. This is the measure of angle D , not the measure of angle F .

Choice C is incorrect. This is the measure of angle E , not the measure of angle F .

Choice D is incorrect. This is the sum of the measures of angles E and F , not the measure of angle F .

Q5**A****A.** 40

Choice A is correct. Given that lines m and n are parallel, the angle marked 130° must be supplementary to the leftmost angle marked a° because they are same-side interior angles. Therefore, $130^\circ + a^\circ = 180^\circ$, which yields $a = 50^\circ$. Lines ℓ and m intersect at a right angle, so lines j , ℓ , and m form a right triangle where the two acute angles are a° and b° . The acute angles of a right triangle are complementary, so $a^\circ + b^\circ = 90^\circ$, which yields $50^\circ + b^\circ = 90^\circ$, and $b = 40$.

Choice B is incorrect. This is the value of a , not b . Choice C is incorrect and may be the result of dividing 130° by 2. Choice D is incorrect and may be the result of multiplying b by 2.

Q6**A****A.** $w = 40$

Choice A is correct. In the figure shown, lines j and k are parallel if and only if a pair of corresponding angles are congruent. It's given that one angle has a measure of 40° and that the corresponding angle has a measure of w° . Therefore, $w = 40$ is sufficient to prove that lines j and k are parallel.

Choice B is incorrect. The angle measuring x° and the angle measuring 40° are alternate interior angles. Thus, if lines j and k are parallel, x is equal to 40, not 140.

Choice C is incorrect. The angle measuring y° and the angle measuring 40° are vertical angles. Thus, $y = 40$, whether lines j and k are parallel or not.

Choice D is incorrect. The angle measuring z° is supplementary to the angle measuring 40° . Thus, $z = 180 - 40$, or $z = 140$, whether lines j and k are parallel or not.

Q7

A

A. 58°

Choice A is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is 180° . It's given that in $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . It follows that the measure of $\angle Z$ is $180 - 24 - 98^\circ$, or 58° .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures of $\angle X$ and $\angle Y$, not the measure of $\angle Z$.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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